

Munkres' General Topology Autoformalized in Isabelle/HOL

85,000+ lines · 806 results · zero sorrys — 24 days of LLM-driven formalization, with no code written by a human

have <...> **sorry**

sledgehammer



have <...> **by (metis ...)**

0

sorrys remaining

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Two months after Megalodon: same book, mature prover

- January 2026: ~130,000 lines of Munkres' topology in ~2 weeks, formalized in Megalodon by ChatGPT 5.2 via Codex — large-scale autoformalization shown feasible, but with significant gaps left.
- The natural question: does a mature system — a large library (Complex_Main), sledgehammer with premise selection and proof reconstruction, and the structured Isar language — make it substantially easier and faster?
- The answer, in practice: both effects. Automation and library support simplify and speed up much of the work — while the richer setting demands more involved oversight of the tools.
- This experiment: the same source material and the same automation loop, in Isabelle/HOL.



THE TIMELINE

Jan 2026

Megalodon experiment: ~130 k lines in ~2 weeks (arXiv:2601.03298)

Mar 2

Phase 1 starts — ChatGPT 5.2 formalizes definitions & statements

Mar 21

Phase 2 starts — Claude Opus 4.6 develops the proofs

Apr 3

Last sorry eliminated — 806 / 806 results fully proved

All of Munkres' point-set topology, fully proved

85,472

lines of Isabelle/HOL
across four theory files

806

lemmas, theorems and corollaries
— every one fully proved

199

definitions, from topologies
to dimension theory

0

sorrys remaining
as of April 3, 2026

24

active days of agent work
(March 2 – April 3, 2026)

39

sections covered — all of
§§12–50, Chapters 2–8



No line of code was written by a human — an intentional design constraint of the experiment. Chapter 1 (set theory and logic) is covered by Isabelle's Complex_Main; everything else was produced by the two LLM agents.

From Tychonoff to a nowhere-differentiable function

- ✓ Urysohn lemma §33.1
- ✓ Urysohn metrization theorem §34.1
- ✓ Tietze extension theorem §35.1
- ✓ Imbedding of manifolds §36
- ✓ Tychonoff theorem §37.3
- ✓ Stone-Čech compactification §38
- ✓ Nagata-Smirnov metrization §40.3
- ✓ Paracompactness (all results) §41
- ✓ Smirnov metrization theorem §42.1
- ✓ Space-filling curve §44.1
- ✓ Ascoli's theorem (classical) §45.4
- ✓ Ascoli, general — via Tychonoff §47.1
- ✓ Baire category theorem (both parts) §48.2
- ✓ Pointwise limits on Baire spaces §48.5
- ✓ A nowhere-differentiable function §49.1
- ✓ Dimension theory & imbedding §50

195 formal results correspond directly to Munkres' numbered statements — covering all 156 non-exercise results in the source (some split into parts).

One LaTeX book in, one Isabelle session out



Source material

- top1.tex — a 7,956-line LaTeX representation of Munkres' Topology, Chapters 2–8 (§§12–50).
- Contains 76 definitions, 109 theorems, 34 lemmas, 13 corollaries, 115 examples.
- Exercises are excluded from the formalization target.



The Isabelle/HOL environment

- Imports Complex_Main only: reals & complex numbers, topological/metric spaces, measure theory, extensive algebra.
- sledgehammer connects Vampire, CVC5 and Z3 — with premise selection and proof reconstruction.
- Plus blast, auto/simp, linarith/presburger, meson/metis.



The decisive contrast with Megalodon: its aby hammer exports to TH0 TPTP with good ATP performance — but has no premise selection and no proof reconstruction. For an LLM workflow that leans on automation, that is a critical limitation.

Two LLMs in sequence, one automated loop



Phase 1 • ChatGPT 5.2 (Codex CLI)

- March 2–13, running continuously on a server — 268 commits.
- Formalized all definitions and theorem statements; proofs mostly deferred via sorry.
- Same model and interface as the Megalodon experiment.



Phase 2 • Claude Opus 4.6 (Claude Code)

- March 21 – April 3, on a notebook with pauses — 884 commits.
- Systematic, theorem-by-theorem proof development.
- Drove the sorry count from 51 down to zero.



The loop: a tmux script watches the session and re-issues a single prompt — *“Read CLAUDE.md and follow instructions”* — whenever the agent finishes or stalls. The human steps in only for focused direction or to correct misbehavior.

The sorry-first workflow

- 1 **Skeleton** — write the declarative proof with sorry at every step — no other tactic permitted in new code
- 2 **Build** — compile to verify the skeleton is well-formed
- 3 **Annotate** — mark every sorry with sledgehammer [timeout = 10]
- 4 **Harvest** — one process_theories run collects all ATP suggestions in a single pass
- 5 **Substitute** — insert the fastest found proof for each step
- 6 **Refine** — where sledgehammer fails, split into finer have blocks and repeat
- 7 **Profile** — time slow tactics with eval_at -t and optimize

THE “ABSOLUTE RULE”

“not by fast, not by linarith, not by simp, not by (rule ...), not even done”

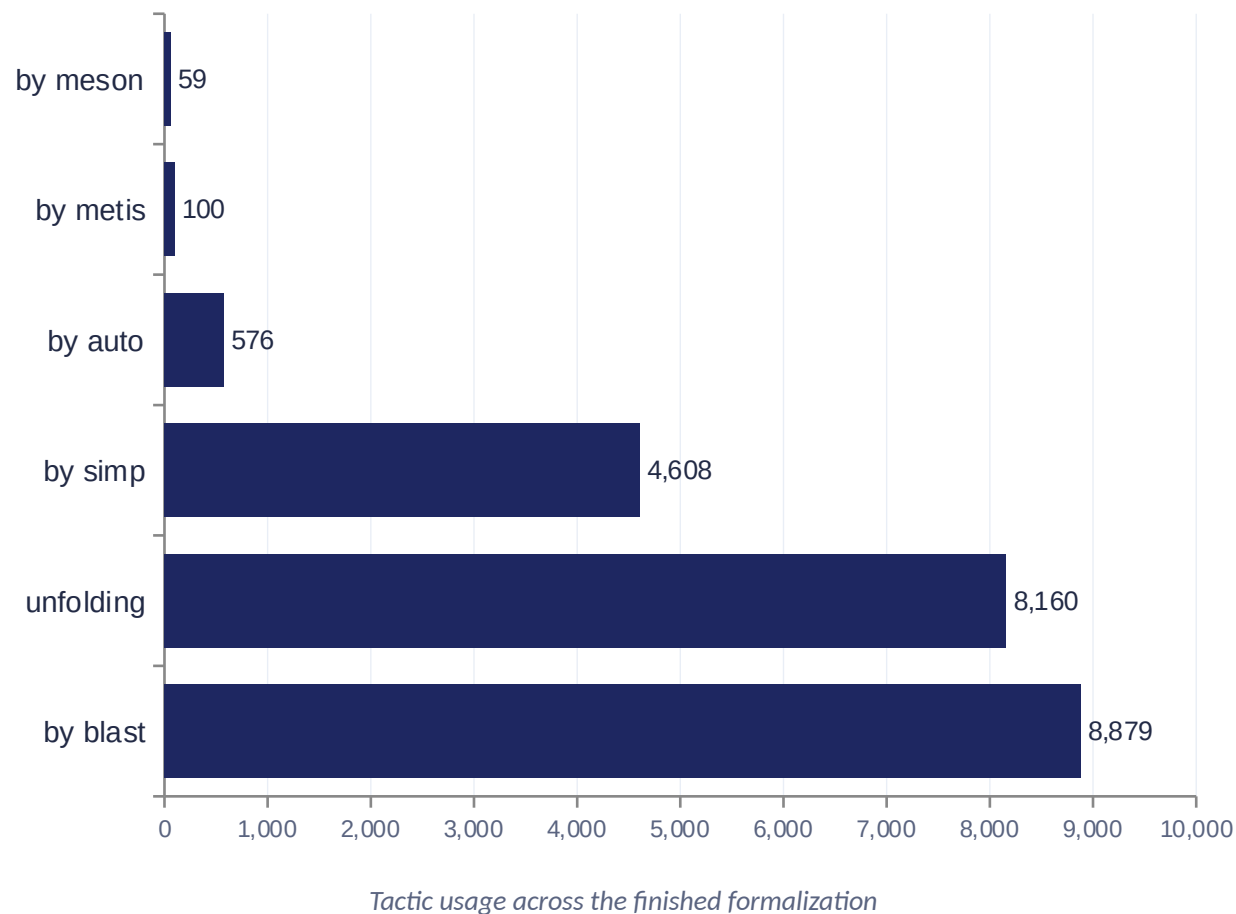
— the instruction file’s final form, hardened on March 24 after repeated tactic explosions



Why so strict?

Complex_Main contributes thousands of simplification rules — one innocent-looking by auto can trigger minutes of fruitless search, stalling the whole loop.

What the discipline looks like at 85,000 lines



- The simpset was never extended: no [simp], [intro] or [elim] attribute anywhere in 85 k lines.
- Every definition is unfolded manually at each use site — 8,160 unfolding calls. Non-standard Isabelle, but immune to cascading simpset explosions.
- In equality-free topology, meson is often ~10× faster than metis — but the LLM lacks the intuition for when to prefer which.



A counterexample to “token prediction is all you need”: an LLM trained on a corpus with the proof methods removed would likely behave much better here.

Explicit carrier sets — the LLM’s own design choice

- No type classes, no locales, no type definitions: a topology is a pair (X, T) with $T \subseteq \wp(X)$ — mirroring Munkres directly. The choice was made by the LLM, independently of the humans.
- The price: it does not interoperate with Isabelle’s type-class topology, which allows only one topology per type.
- Every name carries a `top1_` prefix; results follow Munkres’ numbering as `Theorem_N_M`.
- Every item is annotated with its source location, giving a complete informal-formal correspondence.

```
definition is_topology_on ::
  "'a set  $\Rightarrow$  'a set set  $\Rightarrow$  bool" where
  "is_topology_on X T  $\Leftrightarrow$ 
    {}  $\in$  T  $\wedge$  X  $\in$  T  $\wedge$ 
    ( $\forall U. U \subseteq T \rightarrow \bigcup U \in T$ )  $\wedge$ 
    ( $\forall F. \text{finite } F \wedge F \neq \{\} \wedge F \subseteq T \rightarrow \bigcap F \in T$ )"
```

```
(** from §37 Theorem 37.3 (Tychonoff) [top1.tex:5253] **)
```

the per-item source annotation, as it appears in the theory files

Foreshadowing §8: this definition is also where the review found the most consequential weakness.

85,472 lines across four theory files

File	Sections	Lines	Definitions	Results	Sorrys
Ch2 — spaces & continuity	§§12–22	20,751	78	189	0
Ch3 — connectedness & compactness	§§23–29	13,458	28	107	0
Ch4 — countability & separation	§§30–36	18,908	18	123	0
Ch5–8 — Tychonoff ... dimension theory	§§37–50	32,355	75	387	0
Total	§§12–50	85,472	199	806	0

3.1 : 1

helper-to-Munkres ratio —
611 helper lemmas support the
195 direct correspondences

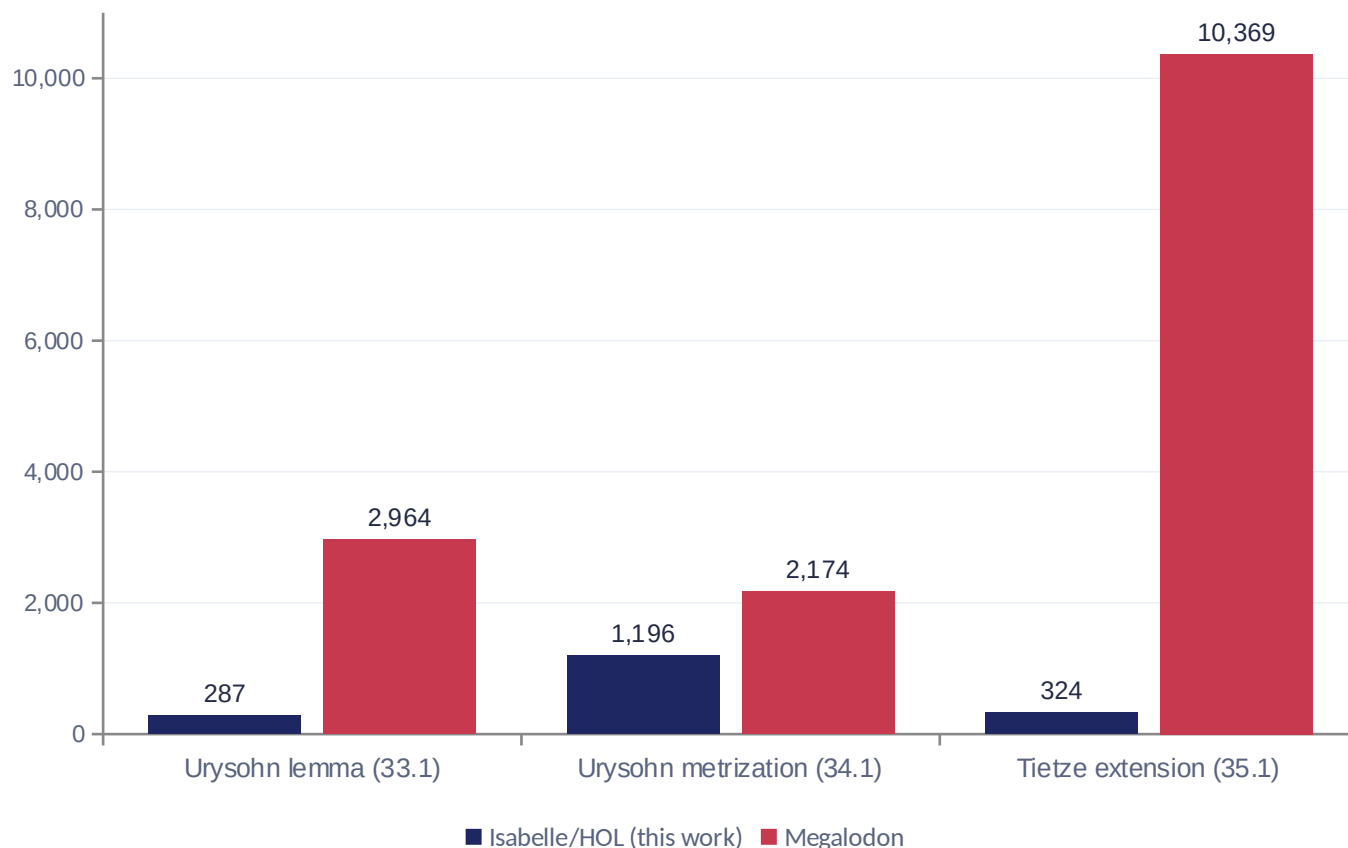
≈ 25

median direct proof length
in lines — typically 2–5
have steps per lemma

156 / 156

non-exercise numbered results
from the book that have
formal counterparts

Order-of-magnitude shorter direct proofs than Megalodon



Direct proof bodies, in lines — same theorems, same book

The “Battle of the Tietze Hill”: ~20 hours and 35 iterations in Megalodon (10,369 lines) — 324 direct lines here.

- `Complex_Main` supplies analytic prerequisites Megalodon must build inline.
- Factoring moves content into helpers: section totals (3,126 / 3,597 lines) sit closer to the Megalodon figures.
- Mulligan–Paulson prove the metrization theorem in 275 lines — riding Isabelle’s type-class library.

Tiny theorems on top of towers of helpers

Theorem	Direct	Section	Helpers
Baire category (48.2)	62	2,907	21
Nowhere-differentiable fn. (49.1)	53	1,296	24
Tychonoff (37.3)	198	1,394	6
Space-filling curve (44.1)	325	1,919	44
Nagata–Smirnov (40.3)	968	2,455	38
Dimension imbedding (50.5)	1,103	6,271	96

Direct = the theorem's own proof body · Section = everything including helpers and definitions

Median direct proof across all 806 results: ≈ 25 lines.



Assembly of pre-verified pieces

The Baire theorem's 62-line proof merely assembles the nested-ball helper and the compact-Hausdorff case; Theorem 49.1's 53 lines assemble 24 lemmas. Each helper was independently discharged by sledgehammer — a direct consequence of sorry-first.

The outlier: the 1,196-line Urysohn metrization proof (34.1) is a monolithic Phase 1 artifact — 260 have steps written before the factoring discipline existed.

Zero sorrys: the five-day final push

355

commits in the final push,
March 30 – April 3

44

commits on April 3 alone
for the space-filling curve

0

sorrys after Corollary 45.5
fell on April 3

THE LAST HOLDOUTS

- \$44** **Space-filling curve** — Hilbert-curve-style recursive definition with snake-order cell traversal; continuity of the limit via uniform convergence.
- \$46** **Sup-uniform topology equality** — identifying the uniform topology on function spaces with its sup-metric description.
- \$47** **Ascoli's theorem, general form** — deeply nested subspace topologies over the compact-convergence topology — depends on Tychonoff.
- \$50** **Dimension-theory imbedding** — general-position arguments, partitions of unity, weighted-sum constructions into $\mathbb{R}^{2^m+1}$.
- 45.5** **The very last sorry** — compact subsets of a complete metric space are exactly the closed and totally bounded ones.

Tychonoff in six helpers

- The first major Phase 2 result — proved March 21 at 07:52, on the first day of proof development.
- Follows Munkres' route: the finite intersection property plus Zorn's lemma.
- Six helper lemmas, including Lemma_37_1 (maximal FIP collections exist, via Zorn), Lemma_37_2 (their closure properties), and `tychonoff_subbasis_in_maxFIP`.
- The recurring pain point: Hilbert's ϵ operator (SOME) — the chosen witness must be shown to satisfy its defining properties across multiple contexts.

```
theorem Theorem_37_3:  
  assumes hIne: "I ≠ {}"  
  assumes hComp: "∀i∈I. top1_compact_on (X i) (TX i)"  
  shows "top1_compact_on (top1_PiE I X)  
        (top1_product_topology_on I X TX)"
```



Why it matters here

Arbitrary products of compact spaces are compact — and the general Ascoli theorem (47.1) later builds directly on this formalization of it.

From Baire to a nowhere-differentiable function

1

Completeness — $C([0,1], \mathbb{R})$ with the sup metric is complete (C01_complete) — hence a Baire space.

2

Open & dense — the difference-quotient sets U_n are open (top1_U49_open) and dense (U49_dense_approx) — density via piecewise-linear “triangle function” approximants.

3

Assemble — the Baire property yields a continuous function on $[0,1]$ that is differentiable nowhere (Theorem 49.1).

Supporting analysis: Heine–Cantor uniform continuity, Archimedean bounds for the mesh size, and careful partition reasoning for the piecewise approximant — a substantial piece of formal analysis sitting atop the topology.

ONE SESSION, ONE THEOREM

14 h

one continuous session
on March 25

40

commits within
that session

24 + 53

supporting lemmas + lines
in the final assembly proof

What 24 days of agent work looks like

Phase 2 (Claude Code) log: 92,524 JSONL records, 1.3 GB compressed — mined with LLM-written analysis tools

50,119

assistant messages
across the session

17,925

Bash commands — 5,186 builds,
3,266 process_theories runs

7,135

file edits, plus
4,962 file reads

2.66 M

output tokens generated
by the model

17.2 B

cache-read tokens — the
context, re-read constantly

626

automated sessions — median
13 min, longest 263 min



\$8,895 at API list prices — ≈ \$160 effective. The experiment ran on a \$200/month subscription; roughly 80% of one month went to this project. Complete textbook formalization is now cheap.

83% automated — and what the other 17% fixed

- Of 764 non-system user messages, 635 (83.1%) were the automated “Read CLAUDE.md” prompt.
- 129 were manual interventions — 59 of them corrections telling the LLM to stop undesirable behavior.
- Correction themes, by frequency: tactic explosion (9), inefficient tool use (6), cherry-picking easy goals (4), resource management (3).
- Each recurring failure became a rule: CLAUDE.md went through eight revisions, and the “ABSOLUTE RULE” was added on March 24 after repeated tactic explosions.



The human voice, verbatim

“how did you again manage to smuggle in uncontrolled slow by calls??? we were through this multiple times.”

“please stop jumping around looking for low hanging fruit; we have to do it all.”

Autonomy: 626 automated sessions, median 13 minutes — the longest ran 263 minutes (4+ hours) without any human input.

Isabelle vs Megalodon on the same book

	Isabelle/HOL (this work)	Megalodon
Logic	Higher-order logic	Higher-order set theory
LLMs	ChatGPT 5.2 + Claude Opus 4.6	ChatGPT 5.2
Library	Complex_Main (extensive)	Set theory + reals (via surreals)
Automation	sledgehammer + blast/auto	aby (TH0, no reconstruction)
Active days	24	~14
Lines	85,472	~130,000
Remaining gaps	0	significant
Sections	§§12–50 (39)	§§12–50 (39, incl. exercises)



Half the length — Isabelle’s library supplies what Megalodon builds from scratch — roughly 2× shorter overall.



Far fewer gaps — sledgehammer’s premise selection and proof reconstruction close steps the aby hammer leaves open.



Division of labor — proof structure from the neural network, proof completion by the ATPs — more effective than asking the LLM to do both.

Same textbook, same automation loop — perhaps the cleanest cross-system comparison of LLM-assisted formalization to date.

Four systems, one chapter of mathematics



HOL Light • Harrison

Same style of approach (setup by Lee and Urban)
— spanning not just topology but an entire probability textbook (Williams, “Probability with Martingales”).



Isabelle/Assistant • Mulligan–Paulson

Amazon’s LLM integration in Isabelle/jEdit:
Urysohn metrization in ≈ 2.5 hours, 275 idiomatic type-class lines — human-assisted and library-integrated.



Naproche/Felix • Hahn–De Lon

The Urysohn lemma autoformalized in a controlled-natural-language proof checker (submitted to ITP 2026).

Megalodon’s virtue is simplicity: one fast checker, transparent proofs, nowhere for the LLM to get lost. **No single approach dominates — an interesting and evolving research area.**

Correct theorems, leaky definitions

- The flagship finding: `is_topology_on` omits the constraint $T \subseteq \text{Pow } X$.
- Root cause: Isabelle's convention $\bigcap \emptyset = \text{UNIV}$ forces the $F \neq \{\}$ clause in the finite-intersection axiom — which silently drops the carrier requirement.
- The weakness propagates: `openin_on` lets “ $\{1\}$ be open in the space $\{0\}$ ”; `finer_than` happily compares arbitrary set families; `neighborhood_of` is unconstrained too.

```
(* formally derivable with the current definition *)
lemma "is_topology_on {0} {{}, {0}, {1}, {0,1}}"
  - despite {1}  $\not\subseteq$  {0}
```



But nothing proved is wrong

Every theorem carries explicit `is_topology_on` assumptions that re-impose the missing constraints at each use site — the proofs are sound.

The definitions are just less reusable than they could be: sufficient for the proofs at hand, with no defensive design. A characteristic LLM failure mode, and one a cleanup pass can fix.

Where the formalization falls short of the book



Fidelity to Munkres

- Theorem 19.1 drops the “all but finitely many factors equal X_α ” clause — the very feature separating product from box topology.
- Theorem 26.7 proves only binary products; Lemma_23_1 proves a different statement than its label.
- Missing entirely: nets, sequential compactness (28.2), the clopen characterization, several counterexamples.



Proof polish

- No global compression pass: wrapper lemmas keep structured Isar proofs where one metis line would do.
- Stray apply...using idioms and duplicate simp-rule warnings — sledgehammer output accepted unfiltered.
- countable is defined three times; redundant “safe” assumptions weaken several statements.

Estimated cleanup to a reusable library: days, not weeks, of expert effort — building on a sound foundation.

...and what is genuinely impressive



Complete coverage — All 156 non-exercise numbered results from Chapters 2–8 have formal counterparts — and all 806 results are fully proved.



Nothing false — Despite the weak definitions, no proved theorem is wrong: explicit assumptions compensate at every point of use.



Debuggable by construction — Short, independently verified helpers make the landmark proofs read as clean assemblies — their logical structure is transparent.



Genuinely deep mathematics — The nowhere-differentiable function, Nagata–Smirnov, the space-filling curve and the dimension-theory imbeddings are far beyond routine exercises.



Full source traceability — Every definition and theorem is annotated with its § and line in the source LaTeX — a complete informal–formal correspondence.

What we learned about LLM-driven formalization



Three lessons

1

Division of labor wins: proof structure from the model, proof completion from the hammers — don't make the LLM do both.

2

Build time is a first-class design constraint: a 120 s timeout forced 115 s → 11.5 s rebuilds, meson-for-metis swaps, and precompiled heap images.

3

Failure modes are systematic, so rules work: tactic explosion, cherry-picking and weak definitions each became a CLAUDE.md rule.



Open issues

1

The richer the setting, the more supervision: Isabelle's power cuts both ways — Megalodon's bare checker gave the LLM less to get lost in.

2

Qualitative gaps remain: leaky definitions, dropped clauses and unpolished proofs still need expert passes.

3

Library reconciliation is untouched: relating the explicit carrier-set world to Isabelle's type-class topology is an obvious next project.

806 theorems, zero sorrys, 24 days, $\approx$ \$160

- A complete autoformalization of Munkres' general topology in a mainstream proof assistant — with no human-written code.
- Sorry-first plus bulk sledgehammer is an effective LLM-Isabelle interface: structure from the model, completion from the ATPs.
- The interaction was $\approx$ 83% automated; the corrections that were needed became durable rules.
- The result is correct but rough — a days-long expert cleanup would turn it into a reusable library.

“It is quite possible that in 2026 we will get most of (reasonably written) math textbooks and papers autoformalized.”

— the Megalodon paper, two months earlier.
This work supports that assessment.

Code github.com/JUrban/isa_top_autoform1

Thank you — questions welcome